

ImQMD Reaction-Class Fractions and z-pol Breakup Profiles

Scope

The y-pol sample is `data/qmdrawdata/qmdrawdata/20260413ypol/`: ^{112}Sn and ^{124}Sn at 190 MeV/u, $\gamma = 0.50 \dots 0.85$, ynp/ypn, RP360. Each `reactiontype.dat` row contributes one ImQMD event. The denominator is the total number of rows in that file. The z-pol Sn112/Sn124 b-discrete files do not carry `reactiontype.dat` labels. For z-pol, this note reports $P_{\text{breakup}}(b) = N_{\text{dbreak}}(b)/N_{\text{header}}(b)$ from `dbreakbXX.dat`; it is not the same observable as the y-pol reaction-class fraction.

y-pol reaction categories

The four labels present in the files are `dkeep`, `break`, `spall`, and `other`. For Sn112 the largest break fraction is 45.1% at g080 ynp; for Sn124 it is 50.5% at g085 ypn.

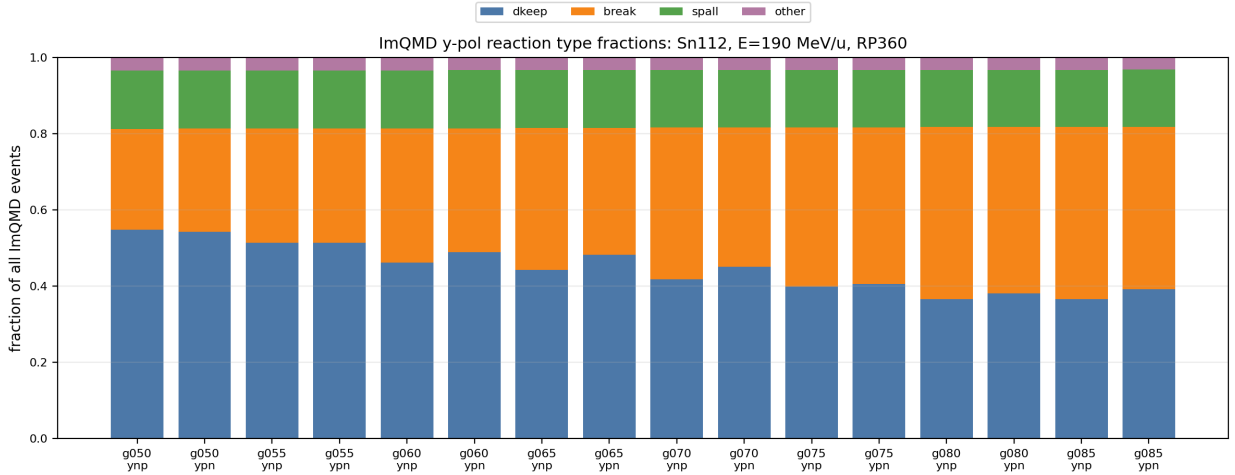


Figure 1: Reaction-type fractions for y-pol ^{112}Sn .

z-pol breakup probability

The largest z-pol b-discrete probability in the scanned Sn112 files is 100.0% at g100 zpn, b=10 fm. For Sn124 the largest value is 99.9% at g100 zpn, b=10 fm.

Artifacts

CSV tables and figures are written under `../.../results/imqmd_reaction_category`.

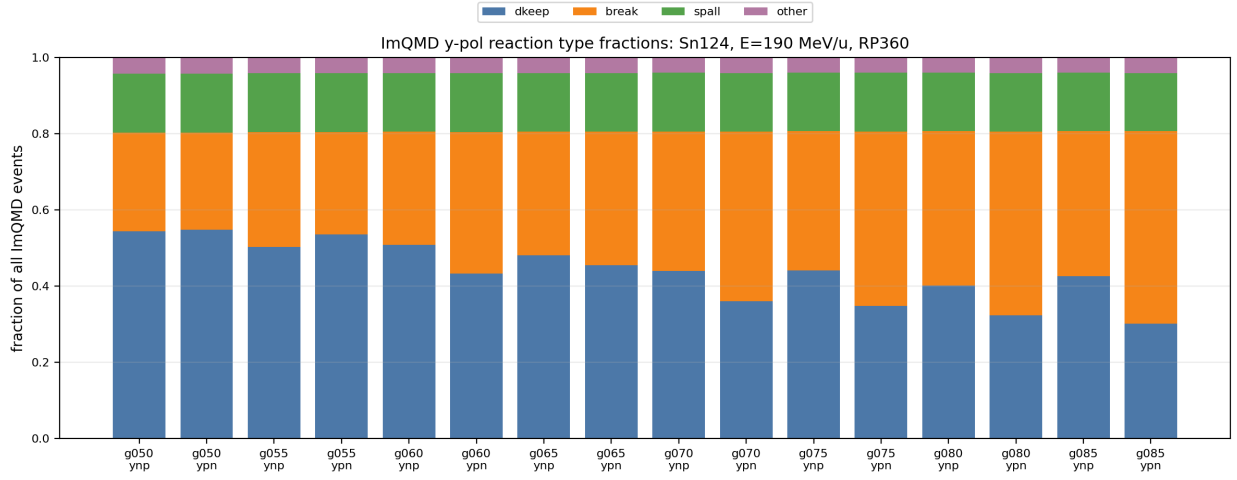


Figure 2: Reaction-type fractions for y-pol ^{124}Sn .

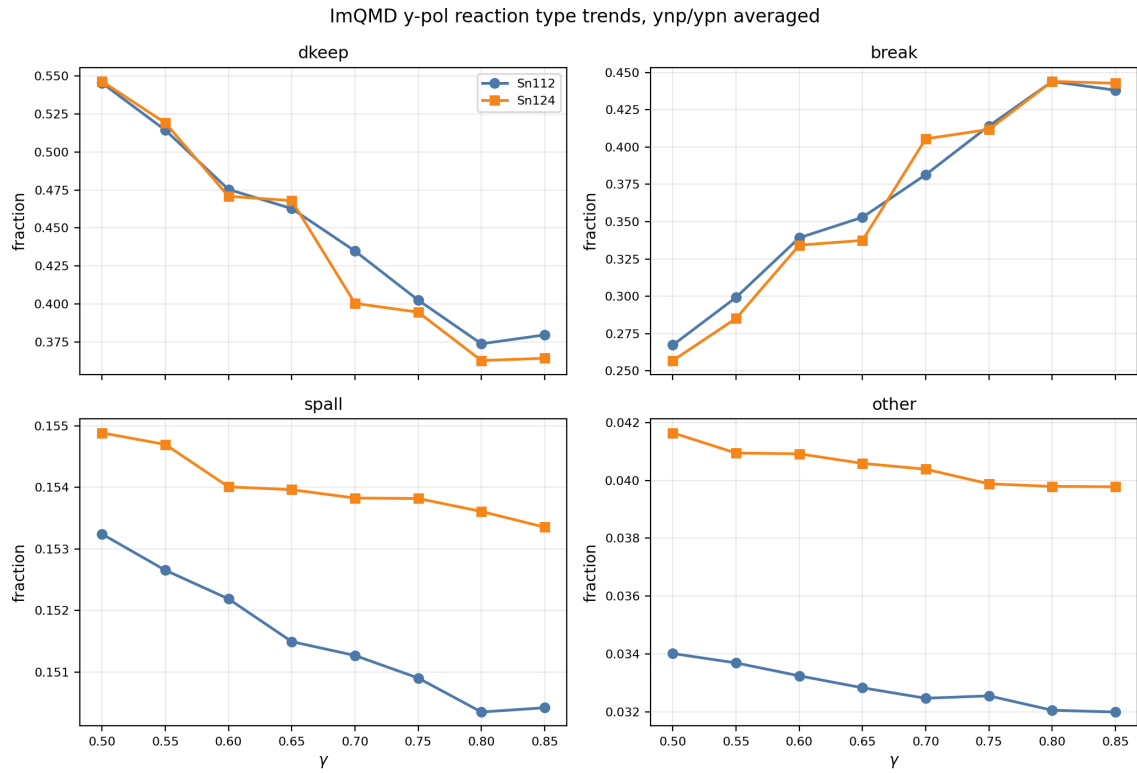


Figure 3: Gamma trends after averaging ynp and ypn at fixed target and gamma.

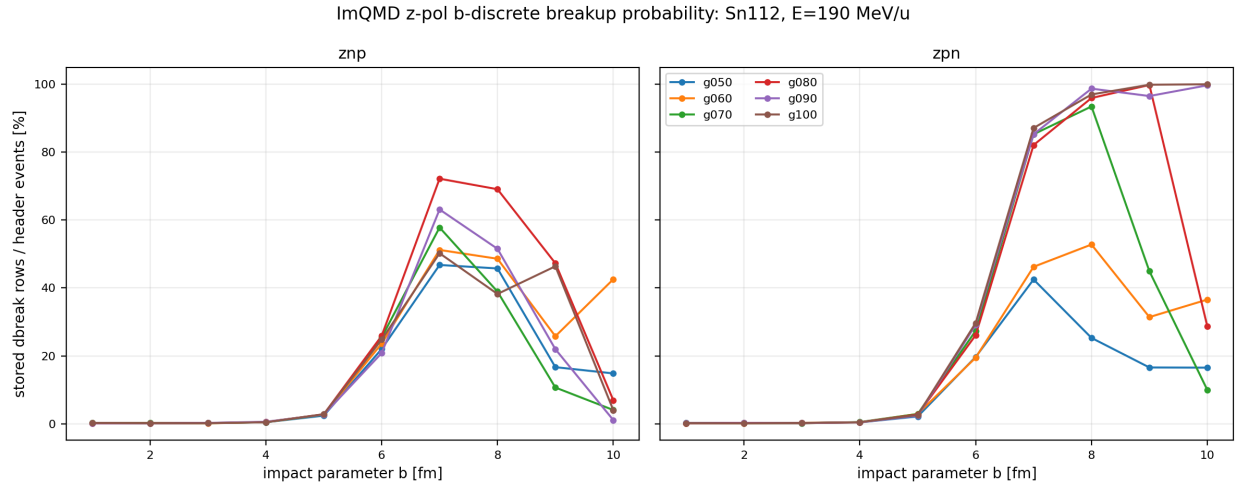


Figure 4: Stored breakup probability profile for z-pol ^{112}Sn b-discrete files.

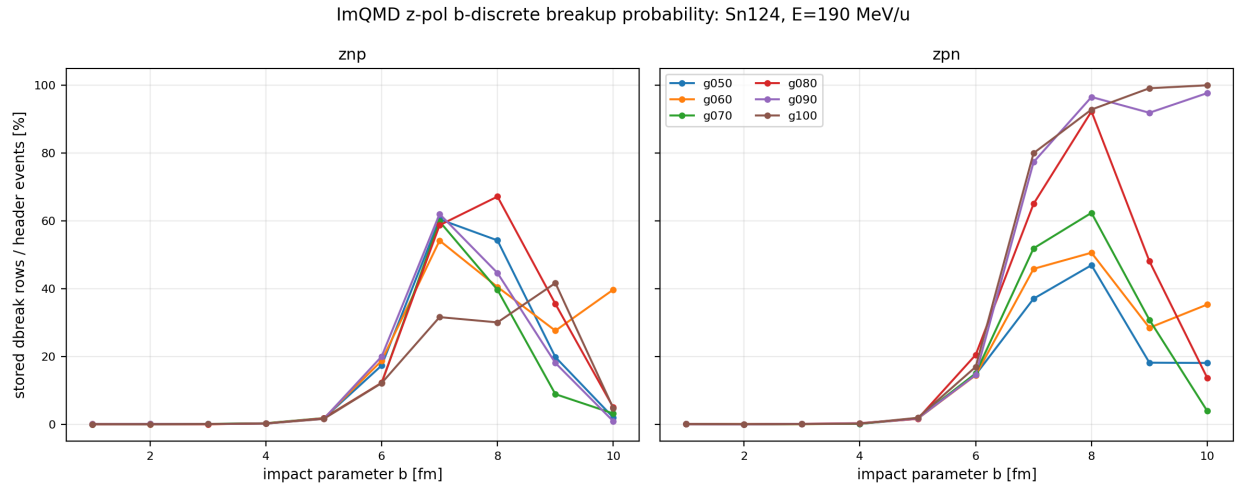


Figure 5: Stored breakup probability profile for z-pol ^{124}Sn b-discrete files.